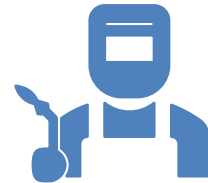


Inside Andy the AI's Brain: How Machines Learn to Decide

- A04_DeepThinkers_Sharise_ITAI2376
- Sharise Griggs, Iman Haamid, Andre Ellis, Erwin Cheng
- Team: Deep Thinkers
- ITAI 2376



Meet Andy

- Andy is a new Artificial Intelligence system learning how to make smart decisions.
- Just like humans, AI studies information before choosing what to do.

What Are Weights?

- Weights tell AI what matters MOST.
- Example:
- When choosing a dog, friendliness may matter more than color.
- ✓ Weights help AI prioritize important information.



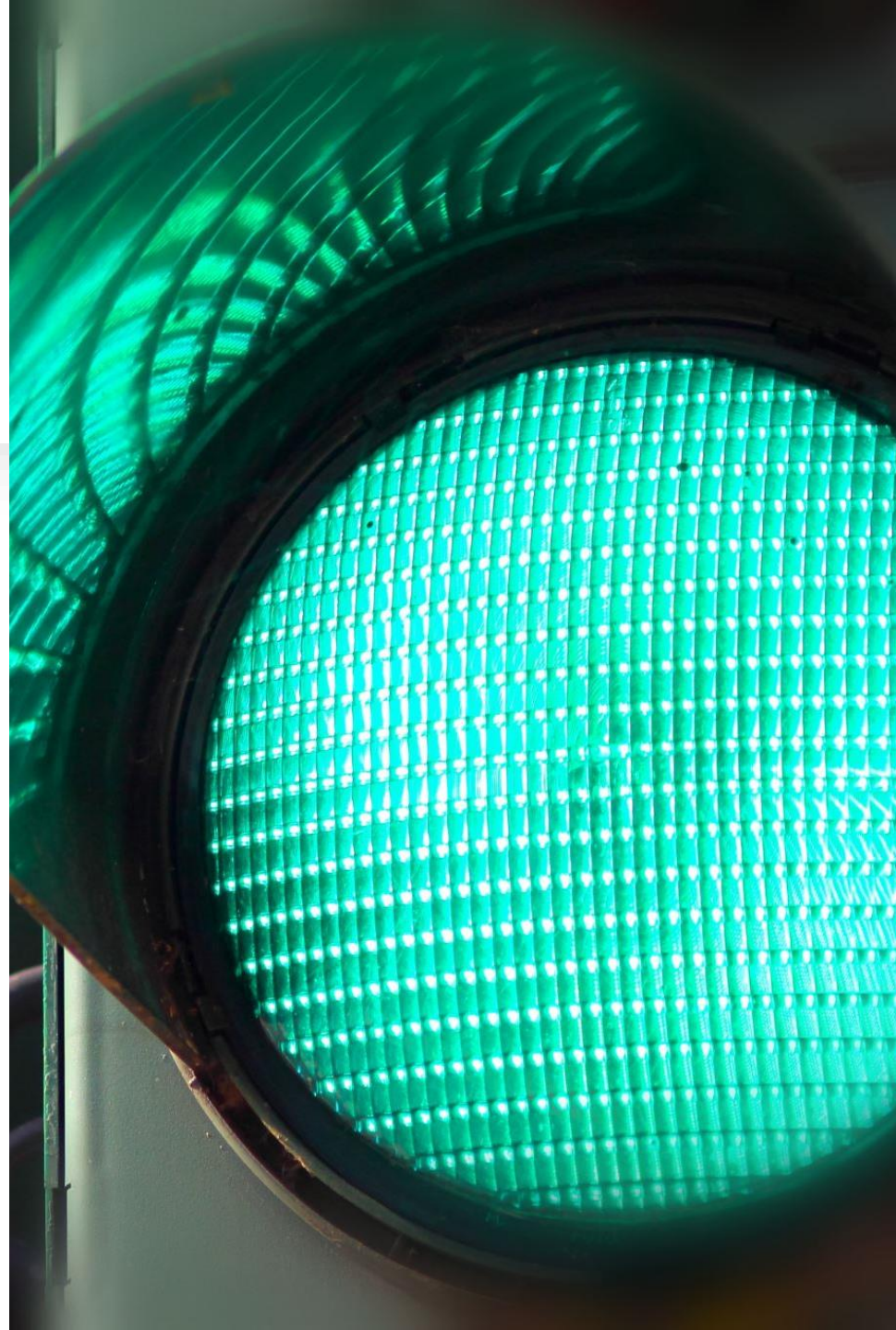
What Is Bias?

- Bias is a starting preference that helps AI lean toward a decision.
- Example:
- If most families want friendly pets, the AI begins to favor friendliness.
- ✓ Bias helps AI make faster, more efficient predictions.



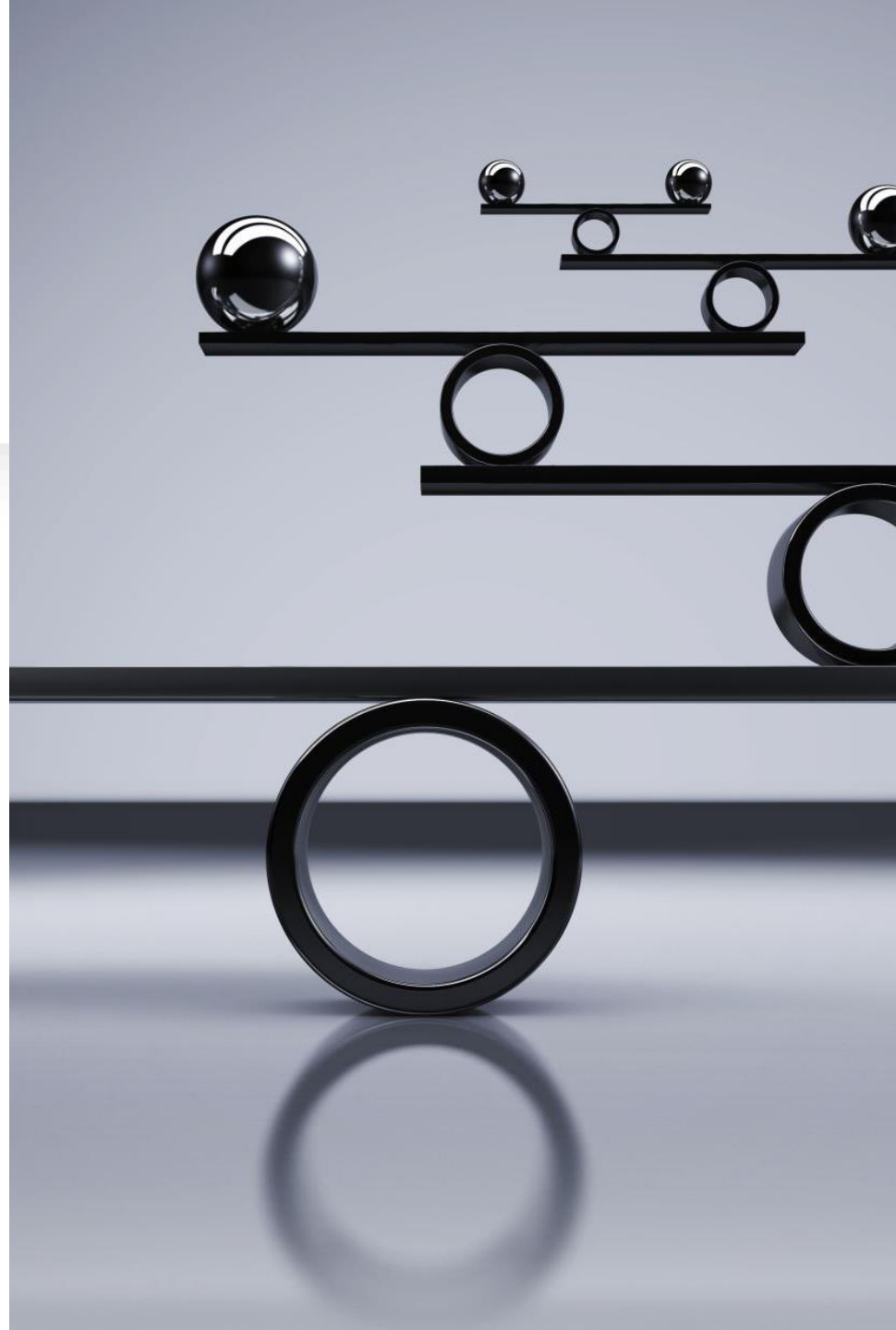
Activation Function: Decision Time

- Activation functions help AI decide whether it should ACT or WAIT.
- Think of a traffic light:
- Green → Go
- Red → Stop
- ✓ Activation functions determine if the signal is strong enough to move forward.



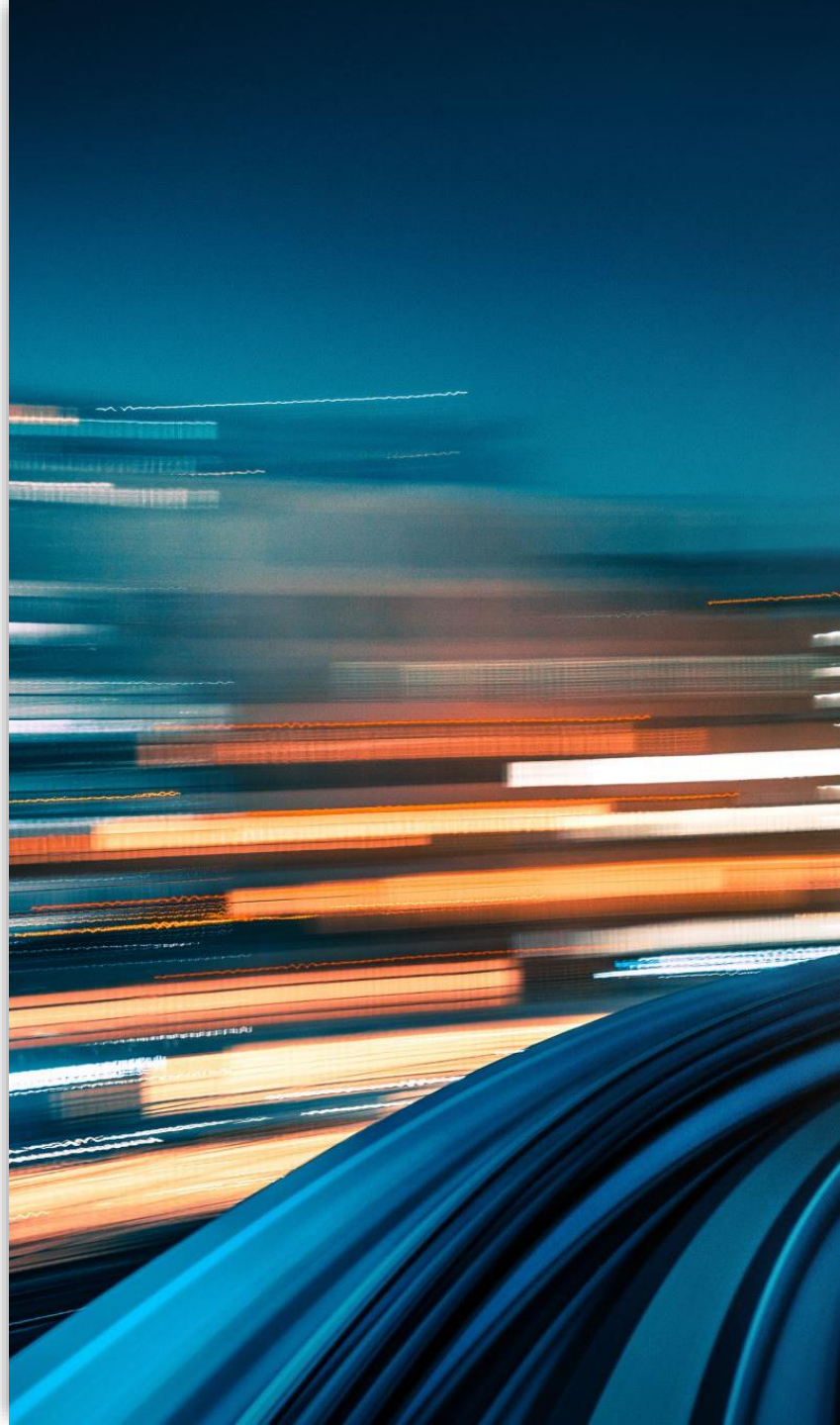
How It All Works Together

- 1. AI studies the data.
- 2. Weights decide what is important.
- 3. Bias provides a starting point.
- 4. Activation determines the final action.
- Together, these components allow AI to make intelligent decisions.



Why This Matters

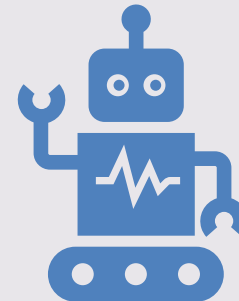
- These concepts power real-world technology:
- Recommendation systems (Netflix, TikTok)
- Self-driving cars
- Healthcare tools
- Fraud detection
- AI is shaping the future of how we live and work.



Final Thought



Weights, bias, and activation functions are the building blocks that allow Artificial Intelligence to learn, think, and make smart decisions.



If an 11-year-old can understand how AI makes decisions, then we are building technology the right way.

Sources

- IBM Cloud Learn Hub — Deep Learning
 - <https://www.ibm.com/topics/deep-learning>
 - Google AI Education
 - <https://ai.google>
 - Microsoft Learn — Introduction to AI
 - <https://learn.microsoft.com>
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